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## ME 353: Electronic Materials and Their Applications

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| <b>Course Code</b>   | ME 353                                      |
| <b>Course Name</b>   | Electronic Materials and Their Applications |
| <b>L-T-P-C</b>       | 3-0-0-3                                     |
| <b>Prerequisites</b> | Consent of the faculty member               |
| <b>Intended for</b>  | B.Tech. 3rd or 4th year                     |
| <b>Distribution</b>  | Core                                        |
| <b>Approval</b>      | 2nd Adhoc Senate (May 2012)                 |

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### Course Contents

- This course covers the advanced aspects of electronic properties of materials and their applications. It includes materials for energy, thermoelectric, ferroelectric, dielectric, pyroelectric, piezoelectric, magnetic and optical applications. The advanced applications of electronic materials in various technologies will be emphasized, Detailed application of these materials, Caloric effect in materials. Linear and non-linear optical properties, materials and applications, Functional composite materials.

### Text and Reference Books

1. A. J. Moulson and J. M. Herbert, **Electroceramics: Materials, Properties and Applications**, 2nd Edition, Wiley; 2003.
2. K. Uchino, **Ferroelectric Devices**, Marcel Dekker Inc., 2000.
3. Z. L. Wang and Z.C. Kang, **Functional and Smart Materials**, Springer, 1998.
4. Charles Kittel, **Introduction To Solid State Physics**, 2nd Edition, 2005.